Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

CREATE Trial Supplemental Appendix

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Supplementary Figures

Figure S1

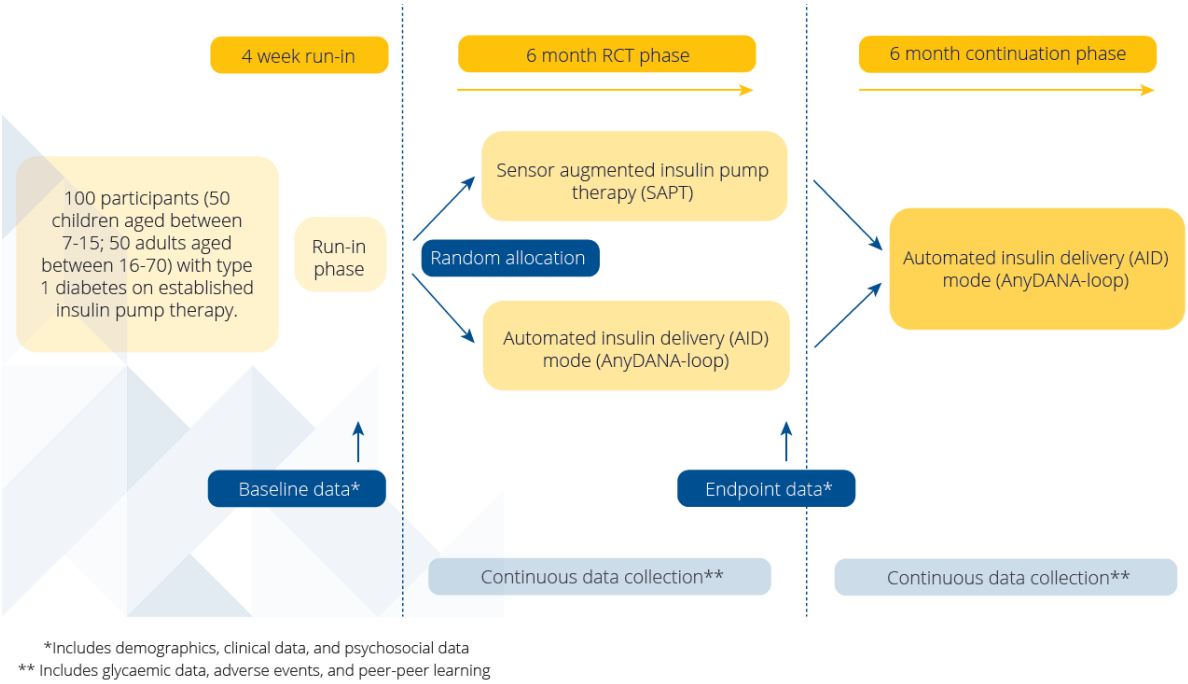

Figure S1: CREATE Trial Study Schema.

Figure S2

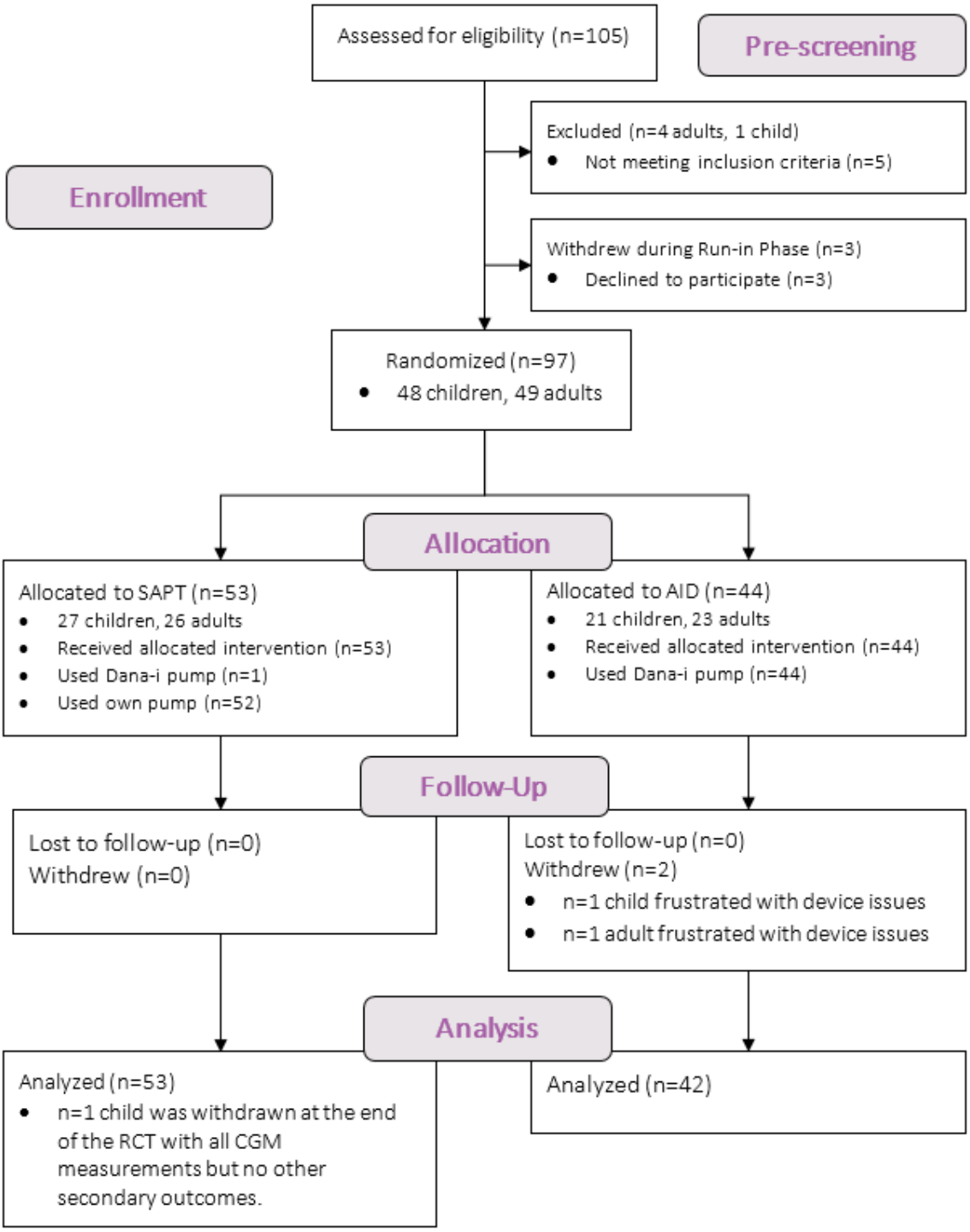

Figure S2: Consort Flow Diagram

Figure S3

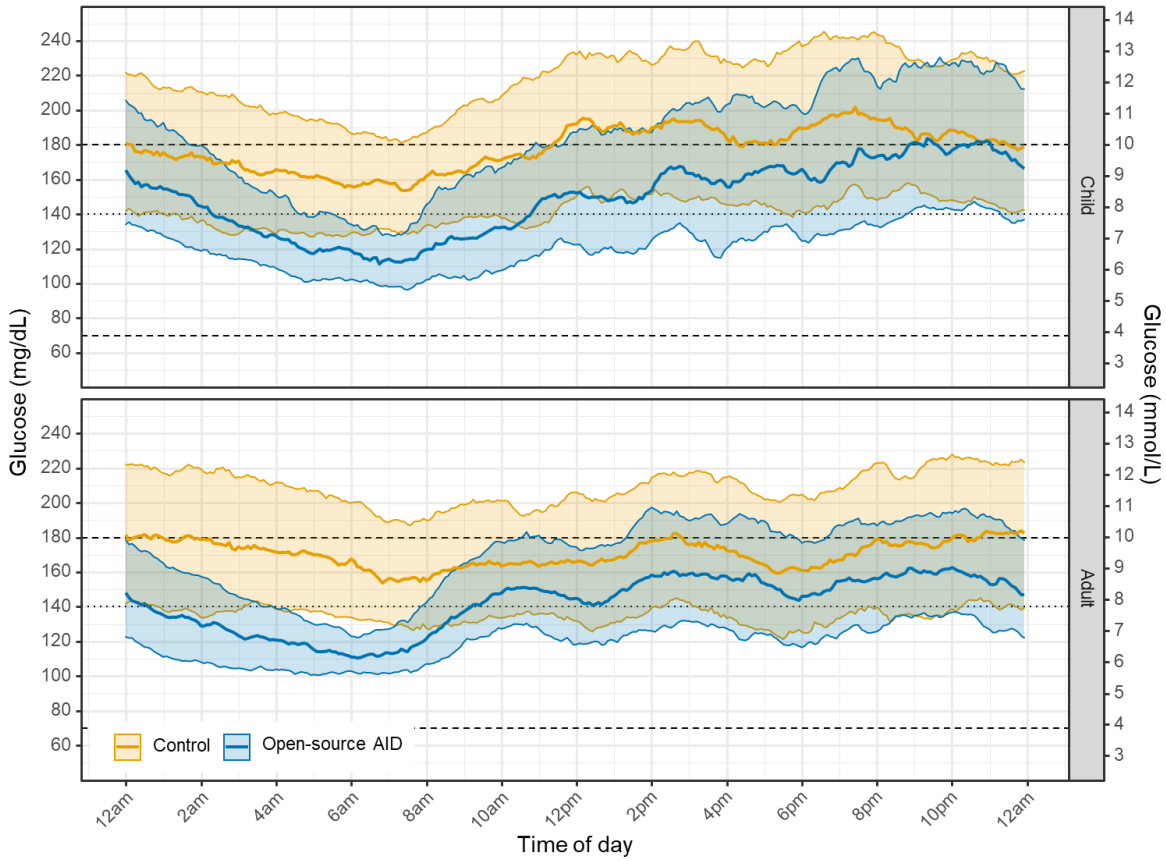

Figure S3: Distribution of Sensor Glucose Values by Time of Day.

Median (solid line) and interquartile range (shaded area) of glucose concentrations, as measured by continuous glucose monitoring during weeks 22 and 23 after randomization, in children (aged 7 to 15 years) and adults (age 16 to 70 years) allocated to either sensor augmented pump therapy (Control, in yellow) or open-source automated insulin delivery (Open-source AID, in blue). Dashed horizontal lines indicate target glucose range (70 - 180mg/dL [3.9 - 10.0 mmol/L]), dotted line indicates lower upper limit of 140mg/dL (7.8mmol/L).

Figure S4

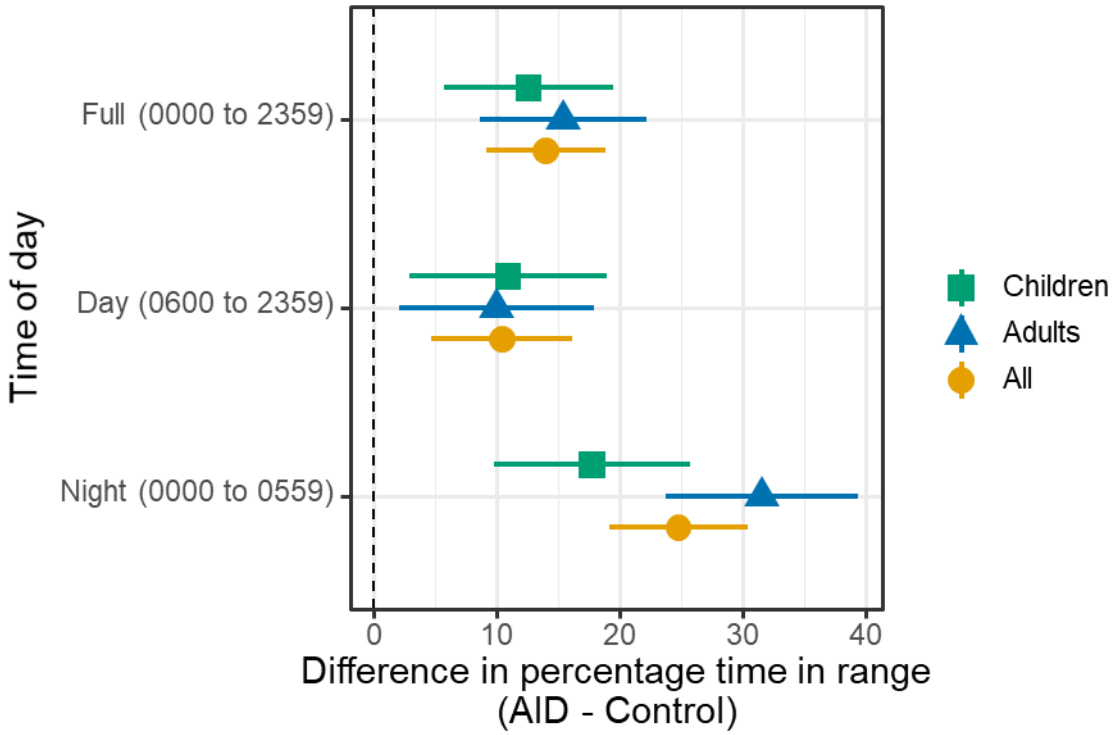

Figure S4: Adjusted Difference in Percentage Time in Target Sensor Glucose Range.

Forest plot of difference in percentage time in target sensor glucose range (70 to 180mg per deciliter, 3.9 to 10.0 mmol per liter), as measured by continuous glucose monitoring during days 155 to 168 (weeks 22 and 23) after randomization, in children (aged 7 to 15 years) and adults (age 16 to 70 years) allocated to either open-source automated insulin delivery (Open-source AID) or sensor augmented pump therapy (Control). Points (horizontal bars) represent estimated differences (95% confidence intervals) between open source AID and control groups, adjusting for stratification variables and percentage time in range at baseline. Dashed vertical line represents null hypothesis (no difference between groups). Confidence interval widths have not been adjusted for multiplicity and may not be used in place of hypothesis testing.

Supplementary Tables

Table S1: Inclusion and Exclusion Criteria for Participation in the CREATE Trial.

Inclusion criteria	Exclusion criteria
• T1D diagnosed as per the American Diabetes Association classification for > 1 year prior to the screening visit. • Aged 7 - 70 years inclusive at baseline. • Currently on insulin pump therapy for > 6 months prior to the screening visit.[a] • Mean HbA1c < 10.5% (91 mmol/mol) within 6 months prior to the screening visit (minimum of one test).[b] • Willing and able to adhere to the study protocol. • Have daily access to a Wi-Fi network.	• If a female is pregnant or plans to become pregnant while participating in the study. A positive urine pregnancy test at screening is exclusionary. • Alcohol or drug dependence. • Severe visual impairment that would impair use of the device. • Any comorbid medical or psychological factors that would, on assessment by the investigators, make the person unsuitable for the study. • A lack of English literacy that would, on assessment by the investigators, make the person unsuitable for the study. • Allergic or intolerant to NovoRapid® insulin.

a) Since carbohydrate counting is a requisite for obtaining a publicly funded insulin pump, the assumption is made that all candidates are capable of carbohydrate counting.

b) This reflects the New Zealand Pharmaceutical Management Agency (PHARMAC) criteria for funded access to insulin pump therapy in New Zealand.

Table S2: Technical Manual for AnyDANA-Loop.

Settings	Recommended Start Settings AID	Adjustment Parameters	Example Scenario
Local Profile (set in Set-up Wizard and accessible from Hamburger Menu)			
DIA (Duration of Insulin Action)	Set to a minimum of 5 hours.	5 - 8 hours. Adjustments can be made throughout the study as per investigator and/or participant discretion. The main reason for extending DIA beyond 5 hours is hypoglycemia 4+ hours after a meal that can't be explained by excessive basal rates or other factors.	A participant is trending below their target glucose level 4+ hours after meals. At least some of these trends result in in hypoglycemia or rescue carbohydrates being administered. As a result, DIA is shifted from 5 hours to 6 hours. Shifting a portion of the insulin action out to 6 hours allows the system to account for the longer action of insulin on glucose levels and reduces hypoglycemia at these time periods.
I:C ratio (Insulin to Carbohydrate Ratio)	Set in negotiation with participant based on run-in data.	I:C ratio defines the number of grams of carbohydrate one unit of insulin covers. Can be adjusted up or down as needed. Postprandial hyperglycemia might suggest a lower ratio is needed. Postprandial hypoglycemia might suggest a higher ratio is needed.	A participant has persistent hyperglycemia after breakfast. Their healthcare professional advises them to shift their I:C ratio from 1:9 to 1:8.
ISF (Insulin Sensitivity Factor)	Set in negotiation with participant based on run-in data.	ISF is a measure of the glucose impact of 1 unit of insulin. If ISF is too low, the system will deliver more insulin than is required to correct high glucose, potentially causing low glucose. If ISF is too high the system will not deliver enough insulin to correct high glucose, leading to extended periods of high glucose.	A participant is spending extended periods of time with high glucose in the evenings. To address this, they reduce their IC ratio from 4 to 3.6.

Basal	Set in negotiation with participant based on run-in data.	Can be adjusted as needed to maintain stable glucose levels when other factors (e.g., food, exercise) are discounted.	A participant notices their glucose levels are consistently above target through the night. They shift their overnight basal from 0.45 to 0.50 to address this.
Target	Set in negotiation with participant based on run-in data. Must be a single number between 5 and 6.5mmol/L.	Can be adjusted within the 5.0-6.5 mmol/L range. The decision about where to set the target is largely down to the size of buffer from low glucose the participant wants from their target range. The lower end of the range is intended for users with excellent control and should not be considered as an ideal target for all participants. Many open-source AID users opt for a target between 5 and 6 (e.g., 5.5mmol/L) and some opt for targets in the 6s. Participants should be encouraged to choose a number they are comfortable with and then adjust other profile settings to optimize time spent at or near their target rather than to change their target glucose to try to improve control.	Participant chooses a target of 6.0 at the start of AID. At their first electronic review, they have had less than 1% time <3.5mmol/L and are happy to change target to 5.5 as their trust in the system has grown.
General			
Language	English. Not to be changed		
Patient name	At set up will be entered as Study ID. Not to be changed.		
General: Protection			
Master Password	Same as other passwords for participant.	Should not be changed unless there is a data security issue. This is the easiest place to change it if needed.	

Settings Protection	None	Can add protection which requires use of biometrics or a password when making settings changes. Should not be used without healthcare professional approval.	
Application Protection	None	Can add protection which requires use of biometrics or a password to access app. Should not be used without healthcare professional approval.	
Bolus protection	None	Can add protection which requires use of biometrics or a password to bolus. Should not be used without healthcare professional approval.	
Home Screen			
Keep Screen on	Off		
Buttons	Use defaults.	This allows the default buttons for the insulin, carbohydrate and calculator to be modified to suit the participant.	Child participant rarely announces carbohydrates of 20g or higher. Carbohydrate increments are changed to 3, 5, 10.
Home screen: Default temp targets			
Activity Duration. This is combined with the Activity Target below)	Default is 120 minutes.	This can be adjusted within the range of 90 – 180 minutes.	Aim is to use activity mode prior to and during activity to reduce risk of hypoglycemia. Longer duration allows for either initiating activity target further in advance of activity, reducing (insulin on board) IOB at outset of exercise or for allowing longer exercise sessions. Example: a road cyclist shifts the activity duration from 120 minutes to 180 minutes, allowing them to more easily set this target for the full duration of their training rides.
Activity Target	Default is 8.0mmolL	This can be adjusted to a value within the range of 7 - 10mmol/l.	The higher the target number, the stronger the signal to AnyDANA-Loop to reduce IOB. Most users will be

			fine at 7.5-8.5, some may find they do not need so strong a signal, or that they need a stronger signal. Example, a runner shifts Activity target to 10.0 so that the system will work to have less IOB prior to their running sessions.
Eating soon Duration. Combined with Eating Soon Target below.	Should be 60 minutes	This can be adjusted within the range of 45 - 60 minutes.	An insulin sensitive participant finds they are at risk of hypoglycemia at/after meal times. To reduce this, they change Eating soon duration to 45 minutes to reduce the active insulin at this point in time.
Eating soon target	Set to target of 4.5 mmol/L.	This can be adjusted to a value within the range of 4.5 - 5.0mmol/l.	
Hypo duration	The default setting is 30 minutes.	This can be adjusted within the range of 30-45 minutes.	Hypo temp targets are set to prevent the system from over-responding to hypo treatments. By setting a higher temp target for a short period after taking fast acting carbohydrate, the system is prevented from delivering too much insulin (and thereby causing a rebound low). If the participant is finding that that the system is responding too strongly to hypo treatments (e.g., with extended increased temp basals), they might increase their duration to 45 minutes to limit the increased basal response (and potential super-microboluses; SMBs) for a longer period of time.
Hypo Target	The default setting is 6.5 mmol/L	This can be adjusted within the range of 6.5-7.5 mmol/L.	Hypo temp targets are set to prevent the system from over-responding to hypo treatments. By setting a higher temp target for a short period after taking fast acting carbohydrate, the system is prevented from delivering too much insulin (and thereby causing a rebound low). If the participant is finding that that the

			system is responding too strongly to hypo treatments (e.g., with extended increased temp basals), they might increase their temp target to as high as 7.5mmol/L to reduce the increased basal response (and potential SMBs).
Fill/Prime standard insulin amounts	Users of Teflon cannulas should have the correct cannula fill entered for button 1. Steel cannula users should change this to 0.	Adjust if participant changes cannula and this changes the cannula fill requirement. Completing the cannula fill on pump leads to this being incorrectly recorded as IOB.	
Range for visualization	3.9-10	Should not be altered as this is related to the primary outcome	
Shorten tab titles	Off	Can be turned on as per participant preference.	
Show notes field in treatment dialogs	Off	Might be of use to children in shared care settings.	
Status lights	Keep at default.	Sets the time after which the age of consumables on the homescreen changes to red. May be altered if participant needs to change set more regularly or if they have issues running out of insulin	
Treatments Safety			
Patient age	Match participant's insulin use to the hard limits associated with specific age categories.		

Age	Bolus Max limit	Max IOB	Max Basal
Child	20	20	5
Adolescent	30	30	10

		Adult	30	30	10
		Insulin resistant adult	40	40	12
Max allowed bolus	Base on bolus data from run and discussion with participant. Should be set high enough to allow for sick day management, high carbohydrate meals, etc.	Can be increased as needed if participant is requiring more bolus insulin. Note: this setting is constrained by the 'patient age' guard rails above.			Teenager is regularly eating high carbohydrate meals and is finding they need more up front insulin. They increase max bolus from 8 to 10.
Max allowed carbohydrates	Base on largest carbohydrate intake during run in or discussion with participant	Max allowed carbohydrates limits the carbohydrates that can be entered into AnyDANA-loop. This cap reduces the risk of user error resulting 500g of carbohydrate being entered instead of 50g. Can be adjusted as needed so that participants' usual carbohydrate intake is well within the limit			Participant has set a max allowed carbohydrates of 90g. At a birthday celebration, they hit the 90gram limit, so adjust the limit to 130 grams in their preferences.
Loop					
APS mode	Open Loop (Stage 1)	Participants can move to from Open Loop TO Predictive Low Glucose Suspend (PLGS Stage 2) after approving 10 recommended basal changes from AnyDANA-loop. Participants can move from PLGS to Closed Loop (Stage 3) after four days of PLGS. Once closed loop has been enabled by progressing through the first two stages, users can access this tab to switch back to open loop or PLGS if needed.			Participants may also choose to go into either open loop or PLGS mode for certain types of exercise activities. Open loop is also useful as a learning tool for participants wanting to understand the system better and may be used when participants are refining their settings. If participants are experiencing problems in stage 3 or 4, they may switch back to PLGS while trouble shooting.

Minimal change request	20%. When in Open Loop, recommendations to approve a new temporary basal rate will only be made when the recommendation exceeds this value from the current rate.	Open loop should only be used at the start of the study and for some users while engaged in prolonged exercise. It is unlikely to need to be changed, and should only be increased.	Participant is using Open Loop during training sessions for an endurance event. They decide they would like fewer recommendations to pop up, so shift from 20% to 30%.
OpenAPS 0.7.0			
Maximum U/hr a temp basal can be set to	This acts as a safety parameter along with the safety multipliers. Can be set to 5x max daily basal to reduce likelihood of needing to increase it as safety multipliers are increased.	If a patient is experiencing significant hypoglycemia, lowering this setting will limit basal corrections. It would usually be preferable to adjust the basal safety multipliers in this scenario, but lowering this setting will work. If a patient is finding that their Max U/hr is limiting AnyDANA-Loop treatment decisions and this is resulting in extended hyperglycemia, they may increase the Max temp basal. This can be identified in the Loop screen in AnyDANA-Loop. Note 1: age bracket (see: Treatments Safety above) places hard guard rails on Max U/hr that cannot be exceeded. Note 2: There are multiple constraints on how much basal delivery can be increased beyond the default setting. These include basal safety multipliers, Max IOB and the maximum basal rate allowed on the pump (set in Dr. Mode menu on the Dana pump).	Participant's max U/hr is set to 4.0. Their highest daily basal rate is 0.9. With initial safety multipliers of 4x current and 3x max daily basal, the 4.0 limit will never be reached. However, if safety caps are increased stepwise from 3x and 4x up to 5x, then the limit of 4.0 u/hr will prevent any increased basal delivery above this number. At this point, the Max u/hr could be increased, or the participant may not be willing to allow an increased basal of this magnitude - in which case, the max u/hr should not be increased.

Maximum IOB	Set as the highest bolus from the previous two weeks + 2x their maximum daily basal or lower as per investigator and participant discretion.	Adjustments can be made throughout the study as per investigator and/or participant discretion. MaxIOB may need to be lowered if someone is having hypoglycemia frequently until other settings are re-calibrated. If a participant is frequently hitting the MaxIOB limit without subsequent hypoglycemia, especially after mealtimes, this can be adjusted to be higher. If a patient is finding that their Max IOB is limiting AnyDANA-Loops treatment decisions and this is resulting in extended hyperglycemia, they may increase the Max IOB. This can be identified in the Loop screen in AnyDANA-Loop. Note: age bracket (see: Treatments Safety above) places hard guard rails on Max IOB that cannot be exceeded.	A participant's highest bolus over 4 week run-in was 10 units. Highest basal rate was 1.2 units MaxIOB could be set at 12.2, or lower. This means if a participant boluses 10u for a meal, the system has room to dose up to 2.2u more of correction. If MaxIOB is set too low, (e.g. MaxIOB is set at 10u per the above example), the closed loop will NOT correct for high BLOOD GLUCOSEs until netIOB lowers below 10u from the bolus decaying. Thus, the MaxIOB should be above the typical bolus amount to allow room for the system to adjust for hyperglycemia. The MaxIOB may need to be adjusted if someone is typically on a lower carbohydrate diet then switches to eating more carbohydrates, therefore requiring higher mealtime and higher correction insulin amounts.
Use Autosens feature	On by default and cannot be turned off.	Only can be adjusted by changing the max and min limits. In closed loop, defaults to 0.80 and 1.2. Defaults adjustment is below. If autosens is not working it indicates a problem with settings. Participants with slow carbohydrate absorption will have fewer autosens changes than those with more rapid carbohydrate absorption	
Enable Supermicrobolus uses (SMB)	Off by default at the start.	If proportion of sensor glucose < 3.5 mmol/L for < 2% in the previous 2 weeks investigators can switch the participant to a more aggressive algorithm (from Oref0 to Oref1) to enable SMB (called stage	SMB is a form of automated bolus correction for treating high blood glucose. SMBs can be delivered in 5 minutes intervals. Each SMB is limited to the smallest of:

		4). Prior to participants making a change to this setting the system will ask them to confirm they have discussed the change with their Healthcare Professional. Once enable SMB has been checked, the participant then needs to also turn on one (or more) of the specific SMB options listed below. These should be turned on in the order they are listed in. Each time a setting is enabled, the participant should ensure they are still meeting the hypoglycemia targets specified above.	– The amount of insulin delivered in 30 min of current basal rate – Half of estimated insulin requirements – The remaining portion of MaxIOB from your settings.
Enable SMB with COB	'Off' by default at the start.	Can be turned on as per protocol (less than 2% time below 3.5mmol/L in the previous 2 weeks).	This is the base SMB setting - it will only operate when there are user-entered carbohydrates (COB = carbohydrates on board) in the app. For SMB to work, both Enable SMB and Enable SMB with COB need to be enabled. When COB is 0, SMB will not be issued. Scenario. At the three week electronic check in, the participant has had 1% time below 3.5mmol/L. They agree with their healthcare professional to enable SMB and Enable SMB with COB, in order to respond to meal related blood glucose excursions more aggressively.
Enable SMB with temp targets.	'Off' by default at the start.	Can be turned on as per protocol (less than 2% time below 3.5mmol/L in the previous 2 weeks). Should only be turned on if Enable with COB has been used for two weeks while maintaining less than 2% time below 3.5	This setting allows SMBs to be issued in response to the "eating soon" temporary target, allowing more aggressive insulin dosing prior to a planned meal. Example, Two weeks after enabling SMB with COB, the participant's time below 3.5mmol/L has remained below 2%. They switch on Enable SMB with Temp Targets to allow more aggressive insulin dosing prior

			to planned meals. Note: This will not enable SMB to run with higher temporary targets (SMBs will only be issued with target blood glucose is below 6.2).
Enable SMB after carbohydrate.	'Off' by default at the start.	Can be turned on as per protocol (less than 2% time below 3.5mmol/L in the previous 2 weeks). Should only be turned on if prior SMB settings have been successfully used without increasing time below 3.5mmol/L.	This setting enables SMB to be issued for up to six hours after carbohydrates are entered (carbohydrates are usually decayed to zero well within 6 hours). This allows SMB to respond to any delayed glucose rises associated with fat and/or protein in meals. Example, two weeks after enabling SMB with temp targets, the participant's time below 3.5mmol/L has remained below 2%. They switch on Enable SMB after Carbohydrate to allow more aggressive insulin dosing for longer after each meal.
Enable SMB always.	'Off' by default at the start.	Can be turned on as per protocol (less than 2% time below 3.5mmol/L in the previous 2 weeks). Should only be turned on if prior SMB settings have been successfully used without increasing time below 3.5mmol/L	This setting enables SMB to operate at all times the target blood glucose is below 6.2. It is the most aggressive SMB setting. Example, two weeks after enabling SMB after carbohydrate, the participant's time below 3.5mmol/L has remained below 2%. They switch on Enable SMB always to allow a more aggressive response to increases in blood glucose.
How frequently SMBs will be given in min	3.	Given that CGM readings are every 5 minutes, in most situations SMBs will be limited to 5 min intervals. In certain unusual circumstances (e.g., entering a calibration for CGM), this will allow SMB to be delivered more frequently. If a user is finding that SMBs are causing lows, they should first review their fundamental settings, then reduce their Max SMB basal minutes before adjusting this setting.	

Max SMB Basal Minutes (Max minutes of basal to limit SMB to).	30 minutes (this setting is only relevant when SMB is enabled).	If proportion of sensor glucose < 3.5 mmol/L for < 2% in the previous 2 weeks investigators and/or participants can increase Max SMB Basal minutes by up to 15 minutes up to a maximum of 60 minutes.	A participant's basal rate is 1u/hr, default of 30 for Max SMB Basal minutes would limit SMB bolus delivery to 0.5u per SMB (subject to other constraints listed above). Shifting to 45 minutes for Max SMB Basal minutes would limit SMB bolus delivery to 0.75.
Max minutes of basal to limit SMB for unannounced meals (UAM)	30 minutes (this setting is only relevant when SMB is enabled).	This setting is combined with the setting above 'Max SMB Basal Minutes' If proportion of sensor glucose < 3.5 mmol/L for < 2% in the previous 2 weeks investigators and/or participants can increase Max minutes of basal to limit SMB to for UAM by up to 15 minutes up to a maximum of 60 minutes.	This setting sets the upper limits of SMB doses for when unannounced carbohydrates are detected.
Enable UAM	Automatically enabled and permanently on.	None	Unannounced Meals (UAM) is a feature that treats rapid rises in blood glucose in the absence of announced carbohydrates as being caused by carbohydrates that have not been entered by the user into the app. It gives users a greater degree of leeway to not enter all carbohydrates, or to make mistakes in estimating carbohydrates (although it is not a replacement for users entering carbohydrates, just a back stop).
Sensitivity raises target	On. When autosens detects increased insulin sensitivity, it will raise the target glucose level to provide buffer from hypoglycemia	Can be turned off if user is finding that this setting regularly causes them to sit at an unnecessarily high glucose level.	Participant finds that their raised target leaves them sitting in the 6.5-7.0 range for hours. They switch the setting off to see if they can sit in the 5.0-6.0 range without dropping into hypoglycemia instead.

Resistance lowers target	Off. When autosens detects increased insulin resistance it applies a lower glucose target to enable increased insulin delivery in response to the insulin resistance	Can be turned on if the user is finding that when autosens detects resistance that their glucose levels stay significantly above their target glucose.	When autosens detects resistance, the user sits at least 1.5mmol above target glucose. They enable the setting to allow additional insulin delivery while resistant.
Minimum carbohydrates required for suggestion	3 grams. When AnyDANA-Loop predicts low glucose, it will post a notification suggesting how many carbohydrates will be required to prevent or treat the low.	Can be adjusted to be higher to reduce the frequency of notifications if the participant finds the recommendations are always needed. Can be adjusted to be lower if the user is finding that threshold for notifying is too high.	Child rarely needs more than 3 grams of rescue carbohydrates. They rarely get notified of need for carbohydrates due to threshold being too high. Threshold is reduced to 2 grams.
OpenAPS 0.7.0 Advanced Settings			
Maximum daily basal safety multiplier	Set as 3x the patient's maximum daily basal rate (units/hour) or per investigator discretion. Note: There are multiple constraints on how much basal delivery can be increased beyond the default setting. These include max basal rate (U/hr) set in AnyDANA-Loop, Max IOB, current	If no significant hypoglycemia (proportion of sensor glucose < 3.5 mmol/L is ≤ 4%), Investigators can shift the highest default basal up a tier (for example, from 3x to 4x or from 4x to 5x). Prior to participants making a change to this setting the system will ask them to confirm they have discussed the change with their healthcare professional. The DANA-I pump has a hard maximum of 500% of current basal rate	This setting is used to make sure that other settings are not disproportionately dosing too much insulin compared to what the body typically needs. This should only be shifted up if there is no significant hypoglycemia and this is noted in the settings as the factor capping the insulin adjustments.

	basal safety multiplier and the maximum basal rate allowed on the pump (set in Dr. Mode Dana pump menu).		
Current basal safety multiplier	Set at 4x the current basal from the participant's run-in period or RCT phase if the patient is switching from SAPT to AID for the continuation phase. Note: There are multiple constraints on how much basal delivery can be increased beyond the default setting. These include max basal rate (U/hr) set in AnyDANA-Loop, Max IOB, maximum daily safety multiplier and the maximum basal rate allowed on the pump (set in Dr. Mode menu).	If no significant hypoglycemia (proportion of sensor glucose < 3.5 mmol/L is ≤ 4%), Investigators can shift the highest default basal up a tier (for example, from 3x to 4x or from 4x to 5x). Prior to participants making a change to this setting the system will ask them to confirm they have discussed the change with their Healthcare Professional. The DANA-I pump has a hard maximum of 500% of current basal rate	Same as above. This setting is used to make sure that other settings are not disproportionately dosing too much insulin compared to what the body typically needs. This should only be shifted up if there is no significant hypoglycemia and this is noted in the settings as the factor capping the insulin adjustments.
Absorption Settings			
Minimum 5 minute	Set to 8.	Investigators can adjust this value to be within the range of 6 - 12, but this is the setting least likely to	This setting drives the decay of carbohydrates when there is no apparent impact on blood glucose from

carbohydrate impact		be adjusted by investigators. Participants should not adjust.	carbohydrate entries. This is not typically used as an adjustment and is not expected to be toggled by investigators. However, it will be possible to adjust it in case there is someone with extreme cases of carbohydrates "never seeming to decay". This setting might be adjusted for a person with gastroparesis.
Meal max absorption time [h]	6	Unlikely to need to be changed. This sets a hard limit after which all carbohydrate from a meal is considered to be absorbed by AnyDANA-Loop.	Unlikely to be changed. May be increased for a person with gastroparesis. Participants following a low carbohydrate diet (so getting a delayed rise in blood glucose) should be advised to follow the recommendations in the Advanced section of the participant guide rather than adjust this setting.
Advanced Absorption Settings			
Autosens ratio	Set at 0.8 to 1.2.	Investigators can change the values to 0.75 to 1.4. Prior to participants making a change to this setting the system will ask them to confirm they have discussed the change with their Healthcare Professional.	Autosensitivity undertakes a rolling assessment of insulin sensitivity over the previous 8 and 24 hours. This produces a sensitivity ratio that is then applied to the algorithm – in short, if it detects that you are more sensitive than usual the ratio will drop from below 1.0, if it detects that you are insulin resistant the ratio will shift above 1 (decreasing or increasing your settings such as insulin to carbohydrate ratio and ISF and thereby altering the amount of insulin delivered). This can be adjusted if someone experiences a lot of sensitivity swings due to exercise, hormonal activity, etc. and the autosens cap is observed as frequently being the limiting factor.
DANA-I			

Selected Pump	n/a. This displays the pump's serial number	This is a simple way to access the pump's serial number. No changes needed unless pump is being changed.	Participant changes pump due to a fault. They tap where the serial number of their current pump is displayed on the app. This opens a pairing window, allowing them to tap the serial number of the new pump to complete the pairing process.
Pump Password	Default is 1234, but users will have chosen at run-in whether or not to enable the pump to be locked.	This allows the password to be viewed. It does not allow the password to changed. This can only be done via the Dr. Mode menu within the pump.	
Bolus speed	12 s/U	Choices are 12/30/60. Participants with problems with insulin delivery such as tunneling around the cannula may choose a slower delivery rate.	Participant has some issues with insulin tunneling. They shift to a 30 sec/U bolus rate and this resolves the problem.
Log reservoir change	On	Automatically logs a reservoir change when participant primes their pump. Can be switched off if participant is performing multiple primes without changing reservoir (although healthcare professional should discuss with participant why multiple primes are needed).	
Log cannula change	On	Automatically logs a cannula change when participant primes their pump. Can be switched off if participant is performing multiple primes without changing cannula (although healthcare professional should discuss with participant why multiple primes are needed).	
Advanced settings DANA-i			
BT watchdog	Off.	When enabled the Bluetooth in the phone is turned off and on again if the phone is unable to connect to the pump. This reduces the chance of a	

		Bluetooth software problem preventing communication between the phone and the pump.	
NS Client			
Nightscout URL	n/a Lists the user's Nightscout URL	Not to be changed unless Nightscout URL changes	
NS API Secret	n/a Shows the users API secret when tapped.	Note: All participants will have a single password, so they can check here if they need their password (e.g., to access their study email account or Dexcom account). No changes in the study unless a data security breach.	
NS Client: Alarm options			
Create notifications from NS alarms	Off	Participants already have access to Dexcom alarms and AnyDANA-Loop alarms, both of which are more useful.	
Create notifications from NS announcements	Off	Participants already have notifications from Dexcom App and AnyDANA-Loop that are more useful.	
Stale data threshold [min]	16 (but not enabled)	Not recommended to be used. This will create an alarm when Nightscout has not received data for the specified number of minutes.	
Urgent stale data threshold [min]	31 (but not enabled)	Not recommended to be used. This will create an additional alarm when Nightscout has not received data for the specified number of minutes.	
NS Client: Connection Settings			
WiFi SSID	None entered	If the user wishes they can enter the SSID number for their trusted wifi connections. AnyDANA-Loop	

		will then not use any other wifi connections to upload to Nightscout (e.g., Hotspot).	
NS Client: Advanced settings			
Autofill BLOOD GLUCOSE	On: This backfills missed readings to Nightscout.	Should be kept on. If a user has a Dexcom problem that results in a large time period where no readings are collected by the Dexcom app, this will likely lead to a gap that cannot be backfilled.	
Create announcements from errors	On: this alerts users to errors with NS client and gives them information to restart NSclient if needed	Should be kept on.	
Create announcements from carbohydrates required alerts	Off. Not needed as AnyDANA-Loop already produces these.		
Local alerts			
Alert if no BLOOD GLUCOSE data is received	On. This provides a notification when CGM data is not being received by AnyDANA-Loop	This data is essential to the study, so needs to be kept on.	
Stale data threshold [min]	30	Ideally won't be changed. If there are regular issues with loss of CGM data, this needs to be dealt with by the study's technical advisor.	
Alert if pump unreachable	On	This should be kept on. If pump is unreachable, AID will not function.	

Alert if carbohydrates are required	On	If the user finds these alerts disruptive, they can be turned off. healthcare professional should discuss with the participant their strategies for avoiding low glucose.	
Pump unreachable threshold [min]	30 minutes	Ideally won't be changed. If a user is regularly losing connection to their pump, this needs to be dealt with by the healthcare professionals initially, then escalated to Hamish/Dana or study's technical advisor.	
Use system notifications for alerts and notifications	On	This means the lock screen will display a notification to the user. It is the most discrete way to alert the user. Keep on if at all possible.	
Data Choices			
Fabric Upload	Default is on	This allows automated crash reporting. It is unlikely, but possible a user may be asked to turn it off when troubleshooting a technical issue.	
Maintenance Settings			
Email recipient	logs@androidaps.org	Would only need to be changed if technical support changed during the trial. Unlikely.	
No of logs to send	2	Only to be changed upon advice from study staff	
Encrypt exported settings	Yes	Would only need to be changed if technical support changed during the trial. Unlikely.	

*Open-source codebase for the CREATE study app available at https://github.com/CREATE-trial/AnyDanaLoop (1).

Table S3: CREATE Trial Schedule of Assessments.

	Screening, Consent & Baseline	Run-in Phase	RCT Phase 24 weeks					Continuation Phase 24 weeks					
		Days 0-28	Days 1 – 168					Days 1 – 168					
	Day 0	Days 0-28 ± 7	Day 1	Day 21 ± 2	Day 42 ± 2	Day 84 ± 7	Day 168 +56	Days 1-42	Day 21 ± 2	Day 42 ± 2	Day 84 ± 7	Days 168 + 7	Days 168-210
Informed Consent	X												
Demographics	X												
Height, Weight, BMI	X		x			x	x				x	x	
Medical Review[A]	X												
HbA1c	X		x			x	x				x	x	
Blood Pressure[B]	X		x			x	x				x	x	
Pregnancy Test[C]	X												
Hypoglycemia Fear Survey II[D]	X						x					x	
PSQI[E]	X						x					x	
DTSQs[F]	X						x					x	
EQ-5D[G]	X						x					x	
Food Diary[H]		x					x						
DANA Pump Training	X												
Dexcom & SAPT Training	X												
AnyDANA-loop Training (participants newly on AID)			x				x						
Pump Setting & Algorithim Review (if applicable)			x			x	x				x	x	
Weekly Electronic Review (all participants)[I]		x											
Electronic Review (participants newly on AID)[I]				x	x				x	x			
Participant Interviews[J]								x					
Interviews (For Māori Participants Only)[K]													x
Pump & Sensor Glucose Data	←											→	
AE Collection & Concomitant Medication Check	←											→	
Health Professionals Focus Group and Interviews[L]			←								→		
Interviews with Participants who Withdrew[M]	←											→	

A. Includes diabetic, clinical and lifestyle review.
B. Seated blood pressure, recorded as the average of two measurements at least five minutes apart.
C. A urine pregnancy test for females of child bearing potential only (all postmenarchal and pre-menopausal women).
D. Versions of the Hypoglycemia Fear Survey II include: Both child and parent version to be completed for participants 7-17 years inclusive, and the adult version to be completed for participants 18 years inclusive and older.
E. The PSQI will only be completed by participants 13 years inclusive and older.
F. DTSQs versions include: Parent version to be completed for participants 7-12 years inclusive, teen version to be completed by participants 13-17 years inclusive (parents can also do this), and the adult version to be completed by participants 18 years inclusive and older.
G. The EQ-5D-Y will be completed by participants 8-15 years inclusive. The EQ-5D-5L will be completed by participants 16 years (inclusive) and older. This questionnaire will not be completed by participants 7 years of age.
H. All participants will be asked to complete a food diary at home using the Research Food Diary app on four non-consecutive days (three weekdays and one weekend day) over one week during the run-in period and during the last week of the RCT phase.
I. Pump settings will be reviewed remotely by investigative staff and adjustments made as clinically indicated.
J. Up to 15 adults and up to 15 children/caregivers, and all Māori participants who have completed the RCT on AID will be invited to attend an interview (face-to-face or via video teleconferencing) with a member of the research team during the first six weeks of the continuation phase (days 1 - 42).
K. All Māori participants who have transferred from the SAPT group to the AID group for the continuation phase will be invited to attend an interview (face-to-face or via video teleconferencing) with a member of the research team within six weeks of completing the continuation phase.
L. One remote focus group will be held with participating HEALTHCARE PROFESSIONALs via video conferencing and will be conducted within one month of all sites having five participants on AID complete the first three months of the RCT phase. Interviews with a selection of HEALTHCARE PROFESSIONALs may also be conducted at each site as all the all participants randomized to AID at their site complete the RCT phase (to occur within one month of the site's last participant completing AID during the RCT phase).
M. All participants who discontinue the trial will be invited to take part in a face-to-face interview (either at the research Centre or via a video conference) with a research staff member within four weeks of discontinuing the study.

Table S4: Sensitivity Analysis of Estimated Adjusted Differences in CGM Metrics and HbA1c.

The primary results were tested for sensitivity to the presence of missing data, and to model changes in glycemic control over time, by entering all CGM data for all participants captured during the RCT phase into a series of longitudinal linear mixed effects models. CGM glycemic metrics were calculated for sequential 14-day (fortnightly) periods, starting 14 days prior to the RCT period and finishing at day 168 (24 weeks). Study time was entered into the models as a fixed effect, where observations recorded during the run-in phase were coded as 0 and fortnightly CGM summaries during the RCT phase as 1,2, … 12. It was expected that the treatment effect may be lower during the first fortnight of the RCT phase while participants adjusted to the new technology, this was captured by adding a three factor variable for study phase coded as 'Run-in', 'RCT fortnight 1', and 'RCT fortnights 2 to 12'. Other fixed effects entered into the models included participants' age strata, and the baseline value of the dependent variable. Interactions were added to allow changes in glycemic control over time to vary by treatment group and age strata. Participant specific changes in glycemic control over time were allowed by adding random intercept and slope effects to the models. An autocorrelation structure of order 1 AR(1) was used to account for possible autocorrelation in patient's values over time, but there was little evidence of this in the data. Maximum likelihood models were fitted using the nlme (version 3.1-152) package in R, and between-group differences at days 155-168 (weeks 22-23) were estimated using the emmeans (version 1.7.0) package.

		AID - Control: Adjusted mean difference at days 155-168		
Time	Glycemic metric	Estimate (95% CI)		
		Child	Adult	Total
24 hours (0000 to 2359)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	12.6 (6.5, 18.7)	15.6 (9.6, 21.6)	14.1 (9.8, 18.4)
	% time in range (3.9-7.8mmol/L [70-140mg/dL])	15.0 (8.9, 21.0)	16.6 (10.6, 22.6)	15.8 (11.5, 20.0)
	% time <3.9 mmol/L [<70mg/dL] (Level 1 or 2 hypo)	0.10 (-0.55, 0.74)	-0.08 (-0.72, 0.55)	0.00 (-0.45, 0.45)
	% time <3.0 mmol/L [<54mg/dL] (Level 2 hypo)	0.02 (-0.23, 0.28)	0.00 (-0.25, 0.25)	0.01 (-0.16, 0.19)
	% time <2.5mmol/L [<45mg/dL]	0.04 (-0.07, 0.14)	0.03 (-0.07, 0.14)	0.04 (-0.04, 0.11)
	% time 10.1-13.9mmol/L [180-250mg/dL] (Level 1 hyper)	-6.4 (-10.1, -2.8)	-9.5 (-13.1, -6.0)	-8.0 (-10.6, -5.4)
	% time 14.0+ mmol/L [250+mg/dL] (Level 2 hyper)	-6.3 (-10.7, -1.9)	-6.0 (-10.3, -1.7)	-6.1 (-9.2, -3.1)

	% time 16.7+ mmol/L [300+mg/dL] (Severe hyper)	-3.7 (-6.7, -0.7)	-2.0 (-4.9, 1.0)	-2.8 (-4.9, -0.7)
	Mean glucose (mmol/L [mg/dL])	-1.3 (-1.9, -0.6) [-22.8 (-34.5, -11.1)]	-1.2 (-1.9, -0.6) [-22.2 (-33.6, -10.7)]	-1.2 (-1.7, -0.8) [-22.4 (-30.6, -14.3)]
	Glucose standard deviation (mmol/L [mg/dL])	-0.25 (-0.57, 0.07) [-4.6 (-10.3, 1.2)]	-0.44 (-0.75, -0.12) [-7.9 (-13.5, -2.2)]	-0.35 (-0.57, -0.12) [-6.2 (-10.3, -2.2)]
	Coefficient of variation (%)	2.4 (-0.0, 4.7)	-0.46 (-2.78, 1.86)	0.93 (-0.76, 2.61)
	Glycated hemoglobin mmol/mol	-4.25 (-7.81, -0.70)	-7.41 (-10.65, -4.16)	-5.71 (-8.09, -3.32)
Day (0600 to 2359)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	9.3 (5.0, 13.6)	11.2 (7.0, 15.5)	10.3 (7.2, 13.3)
	% time in range (3.9-7.8mmol/L [70-140mg/dL])	10.6 (6.6, 14.7)	11.7 (7.7, 15.7)	11.2 (8.3, 14.0)
	% time <3.9 mmol/L [<70mg/dL] (Level 1 or 2 hypo)	-0.39 (-0.84, 0.07)	-0.15 (-0.60, 0.29)	-0.27 (-0.59, 0.05)
	% time <3.0 mmol/L [<54mg/dL] (Level 2 hypo)	-0.14 (-0.30, 0.03)	-0.02 (-0.18, 0.14)	-0.08 (-0.19, 0.04)
	% time <2.5mmol/L [<45mg/dL]	-0.04 (-0.11, 0.03)	-0.02 (-0.08, 0.05)	-0.03 (-0.07, 0.02)
	% time 10.1-13.9mmol/L [180-250mg/dL] (Level 1 hyper)	-4.8 (-7.3, -2.2)	-7.1 (-9.6, -4.6)	-5.9 (-7.7, -4.1)
	% time 14.0+ mmol/L [250+mg/dL] (Level 2 hyper)	-4.1 (-7.1, -1.1)	-4.1 (-7.0, -1.1)	-4.1 (-6.2, -2.0)
	% time 16.7+ mmol/L [300+mg/dL] (Severe hyper)	-2.8 (-4.9, -0.7)	-1.6 (-3.6, 0.5)	-2.2 (-3.6, -0.7)
	Mean glucose (mmol/L [mg/dL])	-0.85 (-1.29, -0.40) [-15.3 (-23.3, -7.2)]	-0.85 (-1.29, -0.41) [-15.3 (-23.3, -7.3)]	-0.85 (-1.16, -0.53) [-15.3 (-20.9, -9.6)]
	Glucose standard deviation (mmol/L [mg/dL])	-0.13 (-0.36, 0.10) [-2.3 (-6.4, 1.7)]	-0.41 (-0.63, -0.19) [-7.4 (-11.4, -3.4)]	-0.27 (-0.43, -0.11) [-4.9 (-7.8, -2.0)]
	Coefficient of variation (%)	1.9 (0.2, 3.5)	-1.4 (-3.0, 0.2)	0.21 (-0.98, 1.41)
Night (0000 to 0559)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	17.8 (12.9, 22.7)	25.4 (20.6, 30.2)	21.7 (18.2, 25.2)
	% time in range (3.9-7.8mmol/L [70-140mg/dL])	22.0 (16.2, 27.9)	28.6 (22.9, 34.4)	25.4 (21.2, 29.5)
	% time <3.9 mmol/L [<70mg/dL] (Level 1 or 2 hypo)	-1.5 (-2.3, -0.7)	-0.00 (-0.79, 0.79)	-0.74 (-1.32, -0.16)

	% time <3.0 mmol/L [<54mg/dL] (Level 2 hypo)	-0.10 (-0.35, 0.15)	0.05 (-0.19, 0.29)	-0.02 (-0.20, 0.15)
	% time <2.5mmol/L [<45mg/dL]	-0.00 (-0.11, 0.11)	0.03 (-0.07, 0.13)	0.01 (-0.06, 0.09)
	% time 10.1-13.9mmol/L [180-250mg/dL] (Level 1 hyper)	-10.6 (-14.2, -6.9)	-16.8 (-20.4, -13.3)	-13.8 (-16.4, -11.1)
	% time 14.0+ mmol/L [250+mg/dL] (Level 2 hyper)	-5.9 (-8.9, -2.9)	-8.5 (-11.4, -5.6)	-7.2 (-9.3, -5.1)
	% time 16.7+ mmol/L [300+mg/dL] (Severe hyper)	-2.2 (-3.7, -0.6)	-2.4 (-3.9, -0.9)	-2.3 (-3.4, -1.2)
	Mean glucose (mmol/L [mg/dL])	-1.4 (-1.9, -0.9) [-25.0 (-33.5, -16.4)]	-2.0 (-2.4, -1.5) [-35.3 (-43.7, -26.9)]	-1.7 (-2.0, -1.3) [-30.2 (-36.3, -24.1)]
	Glucose standard deviation (mmol/L [mg/dL])	-0.52 (-0.79, -0.24) [-9.3 (-14.3, -4.4)]	-0.83 (-1.10, -0.56) [-14.9 (-19.7, -10.0)]	-0.68 (-0.87, -0.48) [-12.2 (-15.7, -8.6)]
	Coefficient of variation (%)	-0.43 (-2.76, 1.90)	-2.6 (-4.9, -0.3)	-1.5 (-3.2, 0.1)

* Confidence interval widths have not been adjusted for multiplicity and may not be used in place of hypothesis testing.
** Mean glucose and glucose standard deviation reported in both mmol/mol with mg/dL in [].
*** Glycated hemoglobin reported in both mmol/mol (SI unit) and derived NGSP units (%).

Table S5: Baseline Participant Characteristics.

Demographics			
Children (< 16 years)	**Control (N=27)**	**AID (N=21)**	**Total (N=48)**
Median age, (IQR) – yr	11.0 (9.0, 14.5)	14.0 (11.0, 15.0)	13.0 (9.0, 15.0)
Gender – no. (%)			
- Female	13 (48%)	11 (52%)	24 (50%)
- Male	14 (52%)	10 (48%)	24 50%)
Ethnicity – no. (%)[a]			
- Maori	4 (15%)	4 (19%)	8 (17%)
- Pacific	0 (0%)	0 (0%)	0 (0%)
- Asian	0 (0%)	1 (5%)	1 (2%)
- European	23 (85%)	16 (76%)	39 (81%)
Body mass index (kg/m^2)- mean (±SD)	19.7±3.1	20.9±4.0	20.3±3.5
Body mass index (kg/m^2)–no.(%)			
- <18.5	11 (41%)	7 (33%)	18 (38%)
- 18.5 to <25	14 (52%)	12 (57%)	26 (54%)
- 25 to <30	2 (7%)	2 (10%)	4 (8%)
Highest qualification (parent) – no. (%)			
- Unknown	7	3	10
- None	0 (0%)	1 (6%)	1 (3%)
- School (1-4)	3 (15%)	4 (22%)	7 (18%)
- Dip. or Cert. (5-6)	6 (30%)	1 (6%)	7 (18%)
- Graduate (7)	6 (30%)	5 (28%)	11 (29%)
- Postgraduate (8-10)	5 (25%)	7 (39%)	12 (32%)
Income (household) – no. (%)[b]			
- Unknown	3	2	5
- < $70,000	3 (12%)	3 (16%)	6 (14%)
- $70,001 - $100,000	7 (29%)	2 (11%)	9 (21%)

- $100,001 - $150,000	4 (17%)	5 (26%)	9 (21%)
- $150,001 or more	10 (42%)	9 (47%)	19 (44%)
New Zealand Deprivation Index (quintile) – no. (%)[c]			
- 1	10 (37%)	9 (43%)	19 (40%)
- 2	10 (37%)	6 (29%)	16 (33%)
- 3	3 (11%)	4 (19%)	7 (15%)
- 4	2 (7%)	2 (10%)	4 (8%)
- 5	2 (7%)	0 (0%)	2 (4%)
Adults (≥ 16 years)	**Control (N=26)**	**AID (N=23)**	**Total (N=49)**
Median age, (IQR) – yr	38.0 (23.2, 45.8)	40.0 (33.0, 43.5)	40.0 (29.0, 45.0)
Age in years, – no. (%)			
- 16 to <25	8 (31%)	2 (9%)	10 (20%)
- 25 to <45	10 (38%)	16 (70%)	26 (53%)
- 45 to <75	8 (31%)	5 (22%)	13 (27%)
Gender – no. (%)			
- Female	15 (58%)	15 (65%)	30 (61%)
- Male	11 (42%)	7 (30%)	18 (37%)
- Non-binary	0 (0%)	1 (4%)	1 (2%)
Ethnicity – no. (%)[a]			
- Maori	5 (19%)	2 (9%)	7 (14%)
- Pacific	0 (0%)	0 (0%)	0 (0%)
- Asian	1 (4%)	0 (0%)	1 (2%)
- MELAA	0 (0%)	1 (4%)	1 (4%)
- European	20 (77%)	20 (87%)	40 (82%)
Body mass index (kg/m^2)- mean (±SD)	27.8±6.4	27.6±5.5	27.7±6.0
Body mass index (kg/m^2)–no.(%)			
- 18.5 to <25	10 (38%)	9 (39%)	19 (39%)
- 25 to <30	9 (35%)	9 (39%)	18 (37%)
- 30+	7 (27%)	5 (22%)	12 (24%)

Highest qualification (patient) – no. (%)			
- Unknown	1	2	3
- School (1-4)	6 (24%)	8 (38%)	14 (30%)
- Dip. or Cert. (5-6)	5 (20%)	1 (5%)	6 (13%)
- Graduate (7)	7 (28%)	6 (29%)	13 (28%)
- Postgraduate (8-10)	7 (28%)	6 (29%)	13 (28%)
Income (household or individual) – no. (%)[b]			
- Unknown	2	2	4
- < $70,000	5 (21%)	3 (14%)	8 (18%)
- $70,001 - $100,000	3 (12%)	2 (10%)	5 (11%)
- $100,001 - $150,000	11 (46%)	5 (24%)	16 (36%)
- $150,001 or more	5 (21%)	11 (52%)	16 (36%)
New Zealand Deprivation Index (quintile) – no. (%)[c]			
- 1	4 (15%)	9 (39%)	13 (27%)
- 2	7 (27%)	3 (13%)	10 (20%)
- 3	7 (27%)	5 (22%)	12 (24%)
- 4	6 (23%)	3 (13%)	9 (18%)
- 5	2 (8%)	3 (13%)	5 (10%)
Smoke: Past vs. Never – no. (%)	1 (4%)	7 (30%)	8 (16%)
Vape: Ever– no. (%)	1 (4%)	0 (0%)	1 (2%)
Diabetes History			
Children	**Control (N=27)**	**AID (N=21)**	**Total (N=48)**
Age at diagnosis, median (IQR)	4.0 (2.0, 6.5)	8.0 (2.0, 10.0)	5.0 (2.0, 8.0)
Years since diagnosis, median (IQR)	6.0 (4.5, 8.0)	5.0 (4.0, 7.0)	6.0 (4.0, 8.0)
HbA1c in prior 6 months mmol/mol [%], mean (SD)[d]	59.0 (9.6) [7.5]	59.8 (8.9) [7.6]	59.3 (9.2) [7.6]

HbA1c at baseline mmol/mol [%], mean (SD)[e]	58.4 (9.9) [7.5]	58.3 (6.6) [7.5]	58.4 (8.5) [7.5]
Severe hypoglycemia episodes[f] - 0 - 1 - 2	 26 (96%) 0 (0%) 1 (4%)	 20 (95%) 1 (5%) 0 (0%)	 46 (96%) 1 (2%) 1 (2%)
Insulin, total daily dose in units, mean (SD)[g]	38.5 (19.4)	46.9 (17.9)	42.1 (19.0)
Prior use of CGM[h]	26 (96%)	20 (95%)	46 (96%)
Prior use of AID[i]	2 (7%)	1 (5%)	3 (6%)
Time in range at baseline (%), mean (SD)[j]	55.1 (12.6)	57.4 (10.6)	56.1 (11.7)
Adults	**Control (N=26)**	**AID (N=23)**	**Total (N=49)**
Age at diagnosis, median (IQR)	14.0 (11.0, 23.8)	14.0 (8.0, 20.0)	14.0 (10.0, 22.0)
Years since diagnosis, median (IQR)	17.5 (10.2, 28.2)	24.0 (16.0, 32.5)	20.0 (14.0, 31.0)
HbA1c in prior 6 months mmol/mol [%], mean (SD)[d]	60.6 (7.0) [7.7]	59.2 (12.7) [7.6]	59.9 (10.0) [7.6]
HbA1c at baseline mmol/mol [%], mean (SD)[e]	62.1 (9.1) [7.8]	60.0 (13.7) [7.6]	61.1 (11.5) [7.7]
Severe hypoglycemia episodes[f]	1 (4%)	0 (0%)	1 (2%)
Insulin, total daily dose in units, mean (SD)[g]	47.7 (13.8)	42.5 (14.5)	45.3 (14.2)
Prior use of CGM[h]	17 (65%)	15 (65%)	32 (65%)
Prior use of AID[i]	5 (19%)	4 (17%)	9 (18%)
Time in range at baseline (%), mean (SD)[j]	60.3 (15.6)	64.7 (12.9)	62.4 (14.4)
Comorbidities			
Children	**Control (N=27)**	**AID (N=21)**	**Total (N=48)**

Other neurological disorders	2 (7%)	0 (0%)	2 (4%)
Chronic pulmonary disease	2 (7%)	3 (14%)	5 (10%)
Adults	**Control (N= 26)**	**AID (N=23)**	**Total (N=49)**
Diabetes, complicated	8 (31%)	12 (52%)	20 (41%)
Cardiovascular pathology	6 (23%)	6 (26%)	12 (24%)
Depression	3 (12%)	2 (9%)	5 (10%)
Other	6 (23%)	2 (9%)	8 (16%)

*Data are presented as n (%) unless otherwise indicated.

[a]Participants could select more than one ethnicity; however, they were allocated to a single ethnic group for statistical output prioritized in the order of Māori, Pacific, Asian and European/Other. For children of non-Māori ethnicity, all identified with European ethnicity and one also identified with Asian ethnicity. All but one Māori child (in the AID arm) also identified with European ethnicity. For adults of non-Māori ethnicity, all identified with European ethnicity, one also identified with Asian ethnicity and one with African ethnicity. All but two Māori adults (both in the control arm) also identified with European ethnicity.

[b]Annual household income in New Zealand dollars.

[c]The New Zealand deprivation index is an area-based measure of socioeconomic deprivation in New Zealand where quintile 5 represents the 20% most deprived areas in New Zealand.

[d]Mean HbA1c (local laboratory values) over the previous six months prior to screening visit.

[e]HbA1c measured using the DCA Vantage Analyzer.

[f]Severe hypoglycemia defined as coma or convulsion requiring assistance from others in the 12 months prior to screening visit.

[g]Total daily dose of insulin calculated as the mean of the previous 14 days.

[h]Prior use of CGM (real time and intermittent) defined as use > 75% of the time prior to the baseline visit.

[i] AID (automated insulin delivery) refers to hybrid closed loop and excludes predictive low glucose suspend and low glucose suspend.

[j]Percentage time with sensor glucose level in target range (3.9-10mmol/L [70-180mg/dL]) during run-in.

Table S6: Representativeness of Study Participants.

Category	Example
Disease under investigation: Type 1 Diabetes (T1D)	
Special considerations related to	
Gender	T1D affects women and men equally (49.9 vs. 50.1%) in New Zealand (2).
Age	Prevalence follows a crescendo decrescendo pattern peaking in the 30-39year age category (18.1%) in New Zealand (2).
Ethnicity	T1D predominantly affects New Zealand Europeans (75.8%) followed by Māori (10.1%), Asian (6.5%), Pacific (4.2%), and Other (1.4%) (2). A more recent audit of children 0-14 years with T1D in New Zealand found 70.2% were European, 18.1% Māori, 7.1% Pacific, and 4.6% Asian. These outputs were prioritized in order of Māori, Pacific, Asian, European/ Other.
Socio-economic status	Patients with T1D in New Zealand are evenly distributed across the five deprivation quintiles (2).
Insulin pump therapy	11.3% of patients with T1D in New Zealand were using insulin pump therapy in 2016, and ethnic minorities have lower rates of insulin pump therapy (2). In 2021, 42.3% of children 0-14 years with T1D in New Zealand were using insulin pump therapy.
Continuous glucose monitoring (CGM)	National rates of CGM use for all patients with T1D in New Zealand are not available; however, 67.2% of children 0-14 years with T1D were using CGM in 2021.
Glycated hemoglobin (HbA1c)	Mean HbA1c for all patients with T1D in New Zealand is unknown; however, mean HbA1c in children 0-14 years with T1D was 67.3mmol/mol [8.3%] in 2021.
Automated Insulin Delivery (AID)	A commercial AID system was found to improve glucose control in patients 7-80 years with T1D in New Zealand (3). Worldwide, no randomized trials have assessed an open-source AID system, and few AID studies report the ethnic proportions of study samples (4).
Overall representativeness of the CREATE trial	Gender was reported by all study participants at baseline; they could report, female, male, or non-binary. The child cohort demonstrated the expected ratio of female to male (1:1); although, the adult cohort had a female predominance (3:2). Study participants ages spanned from 7-70 years, and the majority of patients with T1D in New Zealand fall within this age bracket. The two age bands, children 7-15 years and adults 16-70 years, reflect clinical practice/ service provision in New Zealand for diabetes care. Study participants self-reported ethnicity at baseline, and the study population represented ethnic proportions of people with T1D in New Zealand including a representative sample of Māori (17% of children and 14% of adults). The study was conducted, and hence representative, of the four largest diabetes centers in New Zealand, including two sites in the South Island and two in the North Island. The New Zealand Deprivation is an area-based measure of socioeconomic deprivation in New Zealand where quintile 1 represents the least deprived and quintile 5 represents the most deprived. The study population under-represented the most deprived patients with T1D in New Zealand (quintiles 4 and 5 for children and quintile 5 for adults). Since insulin pump therapy was a requisite for entering the study, the study population had a much higher rate of pump therapy compared to people with T1D in New Zealand. The child cohort in the study were more likely to be using CGM prior to the study compared to the rest of

	children with T1D in New Zealand (96% vs. 67%). The study population also had a lower mean HbA1c compared to the general population with T1D in New Zealand (for children, 59.3mmol/mol [7.6%] vs. 67.3mmol/mol [8.3%]).
Reference (2) data were obtained from the New Zealand Virtual Diabetes Register and are from 2016. Data from children 0-14 years with T1D in New Zealand is available from a 2021 audit (manuscript in draft).	

Table S7: Glycemic Metrics during the Run-In Phase and End of Study, by Treatment Group, Age Strata and Time of Day.

			Control, mean (SD)			AID, mean (SD)			AID - Control: Adjusted mean difference
Age	Time	Glycemic metric	Run-in	Days 155-168	Change	Run-in	Days 155-168	Change	Estimate (95% CI)
Child			N = 27	N = 27	N = 27	N = 21	N = 20	N = 20	
	24 hours (0000 to 2359)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	**55.1 (12.6)**	**52.5 (17.5)**	**-2.6 (-8.1, 2.8)**	**57.4 (10.6)**	**67.5 (11.5)**	**9.9 (2.9, 16.9)**	**12.6 (5.7, 19.5)**
		% time in range (3.9-7.8mmol/L [70 - 140mg/dL])	34.9 (12.7)	31.7 (13.9)	-3.1 (-7.5, 1.2)	35.9 (9.6)	47.4 (11.2)	11.2 (4.2, 18.2)	13.9 (7.5, 20.3)
		% time <3.9 mmol/L [<70mg/dL]	3.7 (3.0)	2.7 (2.8)	-1.0 (-1.9, -0.2)	3.5 (2.6)	2.1 (1.5)	-1.5 (-2.5, -0.4)	-0.5 (-1.6, 0.5)
		% time <3.0 mmol/L [<54mg/dL]	0.7 (0.8)	0.5 (0.7)	-0.2 (-0.5, 0.1)	0.5 (0.5)	0.4 (0.5)	-0.2 (-0.5, 0.1)	-0.1 (-0.4, 0.1)
		% time <2.5 mmol/L [<45mg/dL]	0.2 (0.4)	0.2 (0.2)	-0.1 (0.3)	0.2 (0.2)	0.2 (0.3)	0.0 (0.4)	0.0 (-0.1, 0.2)
		% time 10.1-13.9mmol/L [180-250mg/dL]	25.0 (8.0)	26.0 (7.5)	1.0 (-0.7, 2.8)	25.3 (5.3)	21.1 (6.8)	-4.1 (-8.2, -0.0)	-4.8 (-8.7, -1.0)
		% time 14.0+ mmol/L [250+mg/dL]	16.1 (7.5)	18.8 (14.7)	2.6 (-2.2, 7.4)	13.8 (7.3)	9.3 (6.0)	-4.4 (-8.0, -0.7)	-7.2 (-12.0, -2.4)
		% time 16.7+ mmol/L [300+mg/dL]	6.6 (4.0)	8.1 (10.6)	1.5 (-2.1, 5.1)	6.1 (4.2)	3.6 (3.2)	-2.4 (-4.3, -0.6)	-3.6 (-6.9, -0.4)

		Mean glucose, mmol/L [mg/dL]	9.7 (1.2) [174.1 (22.4)]	10.1 (2.0) [182.4 (36.4)]	0.5 (-2.1, 5.1) [8.3 (-3.4, 20.0)]	9.5 (1.1) [171.0 (19.6)]	8.7 (1.0) [156.8 (18.8)]	-0.8 (-1.4, -0.1) [-13.6 (-25.2, -2.1)]	-1.2 (-1.9, -0.5) [-21.4 (-34.4, -8.4)]
		Glucose standard deviation, mmol/L [mg/dL]	3.9 (0.7) [70.8 (12.2)]	3.8 (0.8) [67.6 (13.6)]	-0.2 (-0.4, 0.1) [-3.2 (-7.4, 0.9)]	3.9 (0.7) [69.6 (12.4)]	3.5 (0.6) [62.2 (10.6)]	-0.4 (-0.7, -0.1) [-7.3 (-13.2, -1.3)]	-0.3 (-0.6, 0.1) [-4.5 (-10.1, 1.0)]
		Coefficient of variation (%)	40.8 (6.1)	37.4 (5.9)	-3.4 (-5.4, -1.4)	40.6 (4.7)	39.7 (5.2)	-0.9 (-3.6, 1.7)	2.1 (-0.4, 4.6)
	Day (0600 to 2359)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	**54.5 (11.9)**	**50.9 (17.4)**	**-3.6 (13.9)**	**56.7 (10.8)**	**64.3 (11.7)**	**7.2 (14.6)**	**10.9 (2.9, 18.9)**
		% time in range (3.9-7.8mmol/L [70 - 140mg/dL])	34.5 (12.2)	30.9 (14.1)	-3.6 (11.3)	35.2 (9.9)	44.0 (10.7)	8.4 (13.7)	11.4 (3.3, 19.6)
		% time <3.9 mmol/L [<70mg/dL]	3.6 (2.7)	2.6 (2.8)	-1.0 (2.3)	3.2 (2.5)	2.4 (1.5)	-0.9 (2.2)	-0.0 (-1.5, 1.4)
		% time <3.0 mmol/L [<54mg/dL]	0.7 (0.7)	0.5 (0.7)	-0.1 (0.6)	0.6 (0.6)	0.4 (0.6)	-0.1 (0.8)	-0.1 (-0.4, 0.2)
		% time <2.5 mmol/L [<45mg/dL]	0.2 (0.2)	0.2 (0.3)	-0.0 (0.2)	0.2 (0.3)	0.2 (0.4)	0.1 (0.5)	0.0 (-0.1, 0.2)
		% time 10.1-13.9mmol/L [180-250mg/dL]	24.7 (7.3)	26.4 (7.9)	1.7 (4.7)	25.8 (5.9)	22.3 (6.3)	-3.3 (8.5)	-4.2 (-9.5, 1.1)
		% time 14.0+ mmol/L [250+mg/dL]	17.2 (7.9)	20.1 (15.2)	3.0 (12.6)	14.2 (8.1)	11.0 (7.0)	-2.9 (8.8)	-6.5 (-11.6, -1.3)
		% time 16.7+ mmol/L [300+mg/dL]	7.3 (4.5)	9.2 (12.1)	1.9 (10.3)	6.3 (4.8)	4.3 (3.8)	-1.9 (4.8)	-3.7 (-6.9, -0.5)
		Mean glucose, mmol/L [mg/dL]	9.8 (1.3) [176.3 (23.1)]	10.3 (2.1) [186.1 (38.6)]	0.5 (1.8) [9.8 (32.3)]	9.6 (1.2) [172.6 (21.9)]	9.0 (1.1) [161.2 (19.6)]	-0.6 (1.4) [-10.4 (25.5)]	-1.1 (-1.9, -0.4) [-20.3 (-34.2, -6.3)]
		Glucose standard deviation, mmol/L [mg/dL]	4.0 (0.7) [72.2 (12.5)]	3.8 (0.7) [69.0 (13.2)]	-0.2 (0.5) [-3.2 (9.6)]	3.9 (0.7) [69.4 (12.8)]	3.6 (0.6) [64.9 (11.2)]	-0.2 (0.8) [-4.2 (13.8)]	-0.1 (-0.5, 0.3) [-2.1 (-9.0, 4.9)]

		Coefficient of variation (%)	41.1 (5.7)	37.5 (5.8)	-3.5 (5.5)	40.2 (5.3)	40.3 (5.3)	0.1 (6.0)	2.9 (-0.3, 6.2)
	Night (0000 to 0559)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	**57.0 (18.9)**	**57.2 (21.4)**	**0.1 (18.9)**	**59.1 (16.5)**	**76.8 (15.8)**	**18.1 (18.6)**	**17.8 (9.8, 25.8)**
		% time in range (3.9-7.8mmol/L [70 - 140mg/dL])	35.8 (16.7)	34.1 (18.7)	-1.7 (16.8)	38.1 (14.6)	56.9 (18.3)	19.4 (24.9)	21.1 (12.9, 29.2)
		% time <3.9 mmol/L [<70mg/dL]	4.2 (4.9)	3.1 (3.5)	-1.1 (3.5)	4.2 (4.6)	1.2 (1.7)	-3.1 (4.8)	-2.0 (-3.4, -0.6)
		% time <3.0 mmol/L [<54mg/dL]	0.9 (2.0)	0.5 (0.7)	-0.4 (1.9)	0.4 (0.4)	0.2 (0.3)	-0.2 (0.5)	-0.3 (-0.6, 0.0)
		% time <2.5 mmol/L [<45mg/dL]	0.3 (1.0)	0.2 (0.3)	-0.2 (0.8)	0.1 (0.3)	0.1 (0.2)	0.0 (0.3)	-0.0 (-0.2, 0.1)
		% time 10.1-13.9mmol/L [180-250mg/dL]	25.8 (12.3)	25.0 (10.9)	-0.8 (10.9)	24.1 (8.4)	17.5 (13.0)	-6.4 (15.5)	-6.6 (-11.9, -1.3)
		% time 14.0+ mmol/L [250+mg/dL]	12.9 (11.3)	14.7 (14.5)	1.8 (13.3)	12.6 (10.8)	4.5 (4.8)	-8.6 (8.7)	-9.2 (-14.4, -4.1)
		% time 16.7+ mmol/L [300+mg/dL]	4.5 (6.2)	4.6 (6.5)	0.1 (8.6)	5.6 (5.9)	1.6 (2.7)	-4.3 (5.0)	-2.6 (-5.8, 0.6)
		Mean glucose, mmol/L [mg/dL]	9.3 (1.60) [167.5 (28.3)]	9.6 (2.0) [171.9 (35.4)]	0.2 (1.6) [4.4 (28.8)]	9.3 (1.4) [166.6 (24.7)]	8.0 (1.3) [144.2 (22.9)]	-1.3 (1.7) [-23.2 (29.8)]	-1.4 (-2.2, -0.6) [-25.5 (-39.4, -11.5)]
		Glucose standard deviation, mmol/L [mg/dL]	3.4 (0.9) [61.9 (16.2)]	3.2 (0.9) [58.4 (16.0)]	-0.2 (1.1) [-3.6 (19.8)]	3.6 (1.1) [65.2 (19.3)]	2.7 (0.7) [48.1 (13.1)]	-1.0 (1.0) [-17.9 (17.3)]	-0.6 (-1.0, -0.2) [-10.9 (-17.9, -4.0)]
		Coefficient of variation (%)	37.2 (8.2)	34.4 (8.2)	-2.8 (8.0)	38.8 (8.6)	33.3 (6.6)	-5.8 (10.2)	-1.9 (-5.1, 1.4)
Adult			**N = 26**	**N = 26**	**N = 26**	**N = 23**	**N = 22**	**N = 22**	

	24 hours (0000 to 2359)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	**60.3 (15.6)**	**56.5 (14.2)**	**-3.8 (-7.7, 0.1)**	**64.7 (12.9)**	**74.5 (11.9)**	**9.6 (4.4, 14.8)**	**15.4 (8.6, 22.1)**
		% time in range (3.9-7.8mmol/L [70 - 140mg/dL])	35.9 (12.7)	32.4 (12.7)	-3.4 (-6.7, -0.2)	41.4 (13.8)	51.1 (14.8)	9.7 (4.0, 15.4)	15.6 (9.3, 21.9)
		% time <3.9 mmol/L [<70mg/dL]	1.7 (2.2)	1.8 (2.5)	0.1 (-0.3, 0.5)	2.3 (2.2)	1.6 (2.1)	-0.4 (-1.2, 0.4)	-0.4 (-1.4, 0.6)
		% time <3.0 mmol/L [<54mg/dL]	0.2 (0.3)	0.3 (0.4)	0.1 (-0.1, 0.2)	0.3 (0.4)	0.2 (0.3)	-0.0 (-0.2, 0.1)	-0.1 (-0.3, 0.2)
		% time <2.5 mmol/L [<45mg/dL]	0.1 (0.1)	0.1 (0.1)	0.0 (0.1)	0.1 (0.2)	0.1 (0.2)	0.0 (0.2)	0.0 (-0.1, 0.1)
		% time 10.1-13.9mmol/L [180-250mg/dL]	26.3 (7.4)	27.9 (8.1)	1.6 (-0.3, 3.5)	24.7 (9.5)	18.2 (8.4)	-6.6 (-11.2, -2.1)	-8.8 (-12.6, -5.1)
		% time 14.0+ mmol/L [250+mg/dL]	11.7 (12.1)	13.8 (9.3)	2.1 (-1.2, 5.4)	8.2 (7.0)	5.6 (4.9)	-2.6 (-5.1, -0.1)	-6.2 (-11.0, -1.5)
		% time 16.7+ mmol/L [300+mg/dL]	4.5 (7.5)	4.9 (5.0)	0.4 (-1.7, 2.4)	2.7 (3.5)	1.8 (2.1)	-0.9 (-2.1, 0.2)	-2.1 (-5.3, 1.1)
		Mean glucose, mmol/L [mg/dL]	9.5 (1.6) [170.3 (28.3)]	9.7 (1.4) [175.4 (24.5)]	0.3 (-0.1, 0.7) [5.1 (-2.5, 12.7)]	8.9 (1.2) [159.7 (21.6)]	8.2 (1.2) [148.4 (21.3)]	-0.7 (-1.1, -0.2) [-12.0 (-20.2, -3.8)]	-1.2 (-1.9, -0.5) [-21.4 (-34.3, -8.6)]
		Glucose standard deviation, mmol/L [mg/dL]	3.3 (0.7) [59.3 (11.9)]	3.4 (0.5) [61.4 (9.5)]	0.1 (-0.1, 0.3) [2.0 (-2.0, 6.1)]	3.2 (0.7) [57.0 (12.9)]	2.8 (0.6) [51.0 (10.7)]	-0.3 (-0.6, -0.1) [-5.6 (-10.1, -1.1)]	-0.5 (-0.8, -0.2) [-8.9 (-14.3, -3.5)]
		Coefficient of variation (%)	35.0 (5.9)	35.3 (5.3)	0.3 (-1.3, 1.9)	35.5 (5.0)	34.1 (3.8)	-0.9 (-3.2, 1.3)	-1.2 (-3.7, 1.2)
	Day (0600 to 2359)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	**59.5 (15.0)**	**57.5 (14.4)**	**-2.0 (9.6)**	**65.4 (12.3)**	**70.9 (12.7)**	**5.3 (11.8)**	**9.9 (2.0, 17.9)**
		% time in range (3.9-7.8mmol/L [70 - 140mg/dL])	35.0 (12.7)	32.9 (12.0)	-2.1 (8.1)	41.5 (13.6)	46.7 (15.6)	5.0 (12.0)	9.9 (1.7, 18.0)

		% time <3.9 mmol/L [<70mg/dL]	1.9 (2.8)	1.8 (2.4)	-0.1 (1.2)	2.4 (2.3)	1.4 (1.6)	-0.6 (1.4)	-0.4 (-1.8, 1.0)
		% time <3.0 mmol/L [<54mg/dL]	0.3 (0.4)	0.3 (0.4)	0.0 (0.3)	0.3 (0.5)	0.2 (0.3)	-0.0 (0.3)	-0.0 (-0.3, 0.2)
		% time <2.5 mmol/L [<45mg/dL]	0.1 (0.1)	0.1 (0.2)	0.0 (0.2)	0.1 (0.2)	0.1 (0.2)	0.0 (0.2)	0.0 (-0.1, 0.1)
		% time 10.1-13.9mmol/L [180-250mg/dL]	26.6 (7.4)	27.2 (7.8)	0.5 (5.2)	24.4 (9.5)	20.8 (8.5)	-3.6 (9.8)	-5.3 (-10.5, -0.1)
		% time 14.0+ mmol/L [250+mg/dL]	12.0 (11.4)	13.6 (9.5)	1.5 (7.5)	7.8 (6.3)	6.8 (5.6)	-1.0 (5.4)	-4.3 (-9.4, 0.7)
		% time 16.7+ mmol/L [300+mg/dL]	4.5 (6.6)	4.8 (4.9)	0.3 (4.3)	2.6 (3.1)	2.2 (2.5)	-0.5 (2.6)	-1.4 (-4.6, 1.7)
		Mean glucose, mmol/L [mg/dL]	9.5 (1.5) [171.3 (26.9)]	9.7 (1.3) [174.6 (24.2)]	0.2 (1.0) [3.3 (17.2)]	8.8 (1.1) [159.0 (20.5)]	8.6 (1.2) [154.1 (22.2)]	-0.3 (1.0) [-5.4 (17.9)]	-0.8 (-1.5, 0.0) [-13.9 (-27.8, -0.1)]
		Glucose standard deviation, mmol/L [mg/dL]	3.3 (0.7) [59.8 (12.6)]	3.4 (0.6) [61.1 (10.4)]	0.1 (0.6) [1.3 (10.1)]	3.2 (0.7) [56.9 (12.2)]	2.9 (0.6) [52.9 (11.1)]	-0.2 (0.5) [-3.7 (9.4)]	-0.4 (-0.7, 0.0) [-6.4 (-13.2, 0.4)]
		Coefficient of variation (%)	35.0 (6.0)	35.2 (5.1)	0.2 (4.5)	35.6 (5.0)	34.1 (3.8)	-1.2 (5.0)	-1.2 (-4.4, 1.9)
	Night (0000 to 0559)	% time in range (3.9 - 10.0 mmol/L [70 - 180mg/dL])	**62.8 (20.9)**	**53.5 (20.1)**	**-9.3 (16.3)**	**62.6 (18.1)**	**85.2 (12.7)**	**22.6 (18.1)**	**31.6 (23.8, 39.4)**
		% time in range (3.9-7.8mmol/L [70 - 140mg/dL])	38.4 (17.4)	30.8 (20.4)	-7.6 (15.3)	40.9 (19.0)	64.4 (15.6)	24.0 (20.6)	32.7 (24.7, 40.7)
		% time <3.9 mmol/L [<70mg/dL]	1.3 (1.5)	1.9 (3.2)	0.6 (3.2)	2.2 (2.4)	2.2 (3.8)	0.3 (3.8)	0.1 (-1.3, 1.5)
		% time <3.0 mmol/L [<54mg/dL]	0.1 (0.2)	0.3 (0.6)	0.2 (0.6)	0.3 (0.5)	0.2 (0.4)	-0.1 (0.6)	-0.1 (-0.4, 0.2)

		% time <2.5 mmol/L [<45mg/dL]	0.0 (0.1)	0.1 (0.2)	0.0 (0.2)	0.1 (0.2)	0.1 (0.2)	0.0 (0.3)	0.0 (-0.1, 0.2)
		% time 10.1-13.9mmol/L [180-250mg/dL]	25.2 (12.0)	30.1 (12.9)	4.9 (12.0)	25.7 (12.4)	10.5 (10.4)	-15.5 (15.6)	-19.9 (-25.0, -14.7)
		% time 14.0+ mmol/L [250+mg/dL]	10.6 (15.5)	14.4 (12.9)	3.8 (13.6)	9.4 (10.5)	2.1 (3.1)	-7.4 (9.4)	-11.7 (-16.7, -6.7)
		% time 16.7+ mmol/L [300+mg/dL]	4.5 (10.6)	5.3 (7.2)	0.8 (9.3)	3.0 (5.5)	0.7 (1.4)	-2.2 (5.4)	-4.1 (-7.2, -1.0)
		Mean glucose, mmol/L [mg/dL]	9.3 (2.0) [167.4 (36.5)]	9.9 (1.9) [178.0 (33.7)]	0.6 (1.7) [10.6 (30.7)]	9.0 (1.6) [161.9 (29.6)]	7.3 (1.2) [131.3 (20.8)]	-1.8 (1.5) [-31.6 (27.8)]	-2.5 (-3.2, -1.7) [-44.5 (-58.1, -30.8)]
		Glucose standard deviation, mmol/L [mg/dL]	3.0 (0.8) [54.2 (13.7)]	3.1 (0.9) [56.2 (16.2)]	0.1 (1.0) [1.9 (18.3)]	3.0 (0.9) [54.8 (16.4)]	2.1 (0.7) [38.5 (11.8)]	-0.9 (0.9) [-15.6 (16.3)]	-1.0 (-1.3, -0.6) [-17.5 (-24.3, -10.7)]
		Coefficient of variation (%)	32.8 (7.2)	31.6 (6.9)	-1.2 (7.3)	33.6 (6.7)	28.9 (5.3)	-3.9 (7.6)	-2.7 (-5.9, 0.5)

*Days 155-168 are the final two weeks of the study and connote "end of study."

**Means and standard deviations are taken from observations made near or after the week 24 assessment, and is missing for 11 participants. The baseline adjusted differences are taken from ordinary least squares regression models.

***CGM metrics and HbA1c were estimated with 95% confidence intervals (CI) using ordinary least squares regression models adjusting for stratification variables (age, study site, and baseline HbA1c > 8.0%/64 mmol/mol) and the baseline level of the dependent variable.

**** Confidence interval widths have not been adjusted for multiplicity and may not be used in place of hypothesis testing.

*****Mean glucose and glucose standard deviation reported in both mmol/mol with mg/dL in [].

******Glycated hemoglobin reported in both mmol/mol (SI unit) and derived NGSP units (%).

Table S8: Subgroup analysis for outcome of CGM-measured % time in range 70-180 mg/dL or 3.9-10 mmol/L.

Baseline factor	Control Arm (Sensor Augmented Pump Therapy)			Intervention Arm (AID)			Absolute difference
		Run-in	Days 155-168		Run-in	Days 155-168	
	N	% mean (SD)	% mean (SD)	N	% mean (SD)	% mean (SD)	Estimate (95% CI)
Predicted TIR							
0% to < 60%	27	48.7 (12.0)	44.3 (13.7)	16	51.3 (5.9)	65.1 (14.4)	20.8 (13.4, 28.2)
60% to 100%	26	67.0 (9.8)	65.0 (10.4)	26	67.6 (11.4)	74.9 (8.7)	9.9 (3.4, 16.4)
HbA1c (mmol/mol) [%]							
<65 mmol/mol [<8%]	35	62.6 (11.7)	61.5 (11.0)	32	64.2 (12.0)	71.3 (10.4)	9.1 (3.8, 14.4)
≥65 mmol/mol [≥8%]	18	48.1 (14.2)	40.8 (15.4)	10	52.6 (10.1)	70.6 (17.1)	27.1 (18.4, 35.7)

* Days 155 to 168 are the final two weeks of the study and represent 'end of study'.

** Estimated differences are adjusted for baseline value of the dependent variable, and stratification variables (age, Hba1c, and clinical center).

*** Predicted time in range (TIR) was estimated for each individual using a linear model with age at consent, gender, baseline time in range, and baseline HbA1c entered as independent variables.

**** Confidence intervals have not been adjusted for multiplicity and may not be used for hypothesis testing.

Table S9: Percent Time in Automated Insulin Delivery.

Approximated percent time in AID	Child (N=21)	Adult (N=23)	Total (N=44)
Mean (SD)	90.0 (7.0)	89.6 (12.9)	89.8 (10.4)
Mean (95% CI)	90.0 (86.8, 93.2)	89.6 (84.0, 95.2)	89.8 (86.7, 93.0)
Median (IQR)	93.8 (84.1, 95.3)	94.4 (89.0, 95.8)	94.2 (87.3, 95.7)

*Time in automated insulin delivery (AID) has been estimated by assuming participants were not on AID if they went longer than one hour without a temporary basal being applied. Each patient's time not on AID was divided by the total amount of time they were in the study between days 14 and 167 of the RCT phase. The first 14 days were excluded from the analysis as participants were getting used to the system.

Table S10: Device Deficiencies by Treatment Arm and Age Strata.

Device deficiency	Children				Adults			
	Control (n=27)		AID (n=21)		Control (n=26)		AID (n=23)	
	Events	Patients n (%)	Events	Patients n (%)	Events	Patients n (%)	Events	Patients n (%)
Connectivity								
Dexcom transmitter to app	1	1 (4)	1	1 (5)	1	1 (4)	3	3 (13)
Monitor/ADL app to pump	1	1 (4)	4	3 (14)	3	3 (12)	3	2 (9)
Other connectivity issue	0	0	1	1 (5)	0	0	8	6 (26)
Dexcom app to Monitor/ADL	0	0	0	0	1	1 (4)	0	0
Hardware								
Autosetter	2	2 (7)	3	3 (14)	5	4 (15)	6	6 (26)
Dexcom G6	3	2 (7)	1	1 (5)	4	1 (4)	2	1 (4)
Pump	2	2 (7)	7	7 (33)	19	14 (54)	19	11 (48)
Other	0	0	1	1 (5)	0	0	0	0
Phone	0	0	1	1 (5)	4	4 (15)	6	6 (26)
Infusion set								
Steel site failure	2	2 (7)	1	1 (5)	0	0	6	4 (17)
Teflon site failure	0	0	0	0	0	0	1	1 (4)
No reported device deficiencies	-	20 (74)	-	7 (33)	-	8 (31)	-	8 (35)

* Follow up days in the randomized controlled trial: Children Control 4227; children AID 3431; adult Control 4368; adult AID 3778.

** Device deficiency is an inadequacy of a medical device with respect to its identity, quality, durability, reliability, safety or performance (requirements of ISO 14155:2011). Device deficiencies with SADE potential were managed as per a SADE. Use errors were not recorded unless they resulted in an ADE.

*** Most device deficiencies related to an insulin pump were attributed to the pre-production Dana-i™ insulin pump battery problem.

References for Supplementary Appendix

1. CREATE-trial/ AnyDanaLoop open-source codebase [Available from: https://github.com/CREATE-trial/AnyDanaLoop].

2. Wheeler BJ, Braund R, Galland B, et al. District health board of residence, ethnicity and socioeconomic status all impact publicly funded insulin pump uptake in New Zealand patients with type 1 diabetes. The New Zealand medical journal. 2019;132(1491):78-89.

3. Collyns OJ, Meier RA, Betts ZL, et al. Improved Glycemic Outcomes With Medtronic MiniMed Advanced Hybrid Closed-Loop Delivery: Results From a Randomized Crossover Trial Comparing Automated Insulin Delivery With Predictive Low Glucose Suspend in People With Type 1 Diabetes. Diabetes Care. 2021;44(4):969.

4. Huyett L, Dassau E, Pinsker JE, Doyle FJ, Kerr D. Minority groups and the artificial pancreas: who is (not) in line? The Lancet Diabetes & Endocrinology. 2016;4(11):880-1.